

EX 9 – Demonstration of Temporal Mining Techniques

AIM :

To demonstrate temporal mining techniques on time-based data to analyze patterns, trends, and sequences over time.

ALGORITHM :

1. Import libraries
2. Load dataset
3. Convert to datetime
4. Sort by time
5. Set time as index
6. Plot time series
7. Decompose into trend, seasonal, residual
8. Apply rolling statistics
9. Detect patterns and anomalies
10. Apply forecasting model
11. Compare predicted vs actual values

CODE :

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("StudentsPerformance.csv")
```

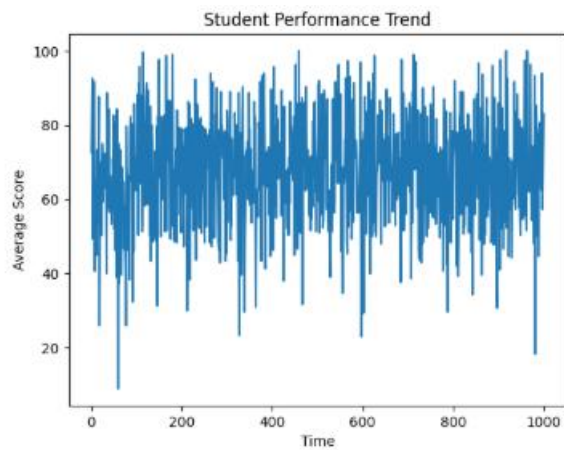
```
df["time"] = range(1, len(df)+1)
```

```
df["average_score"] = (df["math score"] + df["reading score"] + df["writing score"]) / 3
```

```
df = df.sort_values("time")
```

```
plt.plot(df["time"], df["average_score"])  
plt.title("Student Performance Trend")  
plt.xlabel("Time")  
plt.ylabel("Average Score")  
plt.show()
```

OUTPUT :



RESULT :

Thus the above program is executed and the result is obtained.